

Pavanbabu Arjunamahanthi

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Education

PhD in Electrical and Communications Engineering, University of Florida, Gainesville, FL, Jan 2025 – Dec 2028, GPA: 3.62/4

MS in Materials Science and Engineering, University of Florida, Gainesville, FL, Aug 2023 – Dec 2024, GPA: 3.51/4

Work Experience

University of Florida, Graduate Assistant, Gainesville, FL

Jan 2025 – Present

- Engineered sub-5 micrometer hybrid bonding cross-sectional imaging by using targeted ultrasonication and HF Etching, without FIB delayering, improving SEM Quality, reducing charging and contamination artifacts by 99%, and cutting preparation cost by over 50%.
- Developed a new sub-surface interfacial taxonomy with 20+ hybrid bonding manufacturing defects for AI-driven metrology and failure analysis, correlated interfacial defects to warpage behavior, for yield improvement, and process development.
- Accomplished a SAM-based microbump imaging approach by automating the testing process, segmenting, and statistically analyzing over 20000 microbumps across 450 SAM images taken in 8 Advanced packages, saving 40% project time.
- Optimized imaging parameters, improved detection sensitivity and resolution of SAM for voids and delamination in advanced packages with Physics-Informed Neural Network models, and identified the root cause, improving >35% SSIN and >25% PCRS scores.
- Fabricated and measured ALD deposition of thin films in Through Glass Vias (TGVs) and conducted strategic characterization with X-ray and SEM Inspection, identified failure modes, and reduced recurring deposition failures by 40% in the next iterations.
- Achieved 100% non-destructive inspection goals with X-ray, SAM, and Optical Microscopy of fabricated MEMS Devices by proving resolution limitations towards reverse engineering, contributing to secure and reliable Defense-grade electronics.

Applied Materials, Process Engineering Intern, Santa Clara, CA

May 2026 – Aug 2025

- Created a standardized, automated AI-Based Defect detection workflow for μ -LED Wafer defect detection of μ -LED wafers, ensuring precise defect detection and contributing to yield enhancement up to 99.999%.
- Designed AI-optimized heatsink structures for direct cooling of XPU's for maximum thermal efficiency

Indian Institute of Technology, BHU, Project Intern, Varanasi, India

Dec 2022 – Feb 2023

- Achieved a 100% material purity of high-entropy alloy MnBi nanocomposite particle synthesis doped with FeNi@C via Cryogenic Ball Milling and demonstrated a 30% improvement in their magnetic behavior compared to rare-earth magnetic nanoparticles.
- Demonstrated a nanocomposite-based Li-battery anode by confirming the effect of carbon coating of nanocomposite particles for decreased impedance, improved capacitance, and longer cycle life.

Projects

University of Florida, Graduate Assistant, Semiconductor Device Fabrication Project

Jan 2025 – May 2025

- Fabricated a PN junction Diode and a Bi-Morph MEMS Actuator by literature review and quality parameter optimization on a Wafer using photolithography, wet etching, dry etching, Diffusion, Doping, Evaporation, and Plasma Cleaning, saving 20% cost.
- Demonstrated hands-on working ability in thin film deposition using PECVD, ALD, and metrology tools such as Ellipsometry, Profilometry, and SEM Analysis, with experience in processing wafers and reducing iterative process time by 30%.
- Conducted device characterization for IV/CV analysis with a Semiconductor Parameter Analyzer, Bandwidth Measurement, 4 pt probe Measurement, and ensured 70% working device target by careful fabrication.

University of Florida, Graduate Student Assistant, Advanced Packaging Anti-Probing Project

Jan 2024 – Aug 2024

- Led material characterization work on 2.5D/3D advanced packaging interconnects, found 10+ material-related failures and probing regions in 4+ package components and 10+package interfaces, and developed strong probing countermeasures.
- Validated probing attack regions with X-ray tomography, SEM, SAM, FTIR, AFM, and identified underlying defects in thin film regions, Cu Seed layers, RDL Cracks, and Moisture-induced defects by submitting detailed reports and presentations.

Skills

- Characterization:** Atomic Force Microscopy, XRD, X-ray Microscopy, nanoindentation, Scanning Acoustic Microscopy, SEM, Focused Ion Beam, EDS, Dilatometer, XPS, XRF, Raman Spectroscopy, FTIR, TGA, DSC, Ellipsometry, WLI, UV-Vis Spectroscopy, Profilometer, IV/CV Measurement, 4 pt probe Measurement
- Fabrication:** Photolithography, Deep Reactive Ion Etching, Wet and Dry Etching, Evaporation, Plasma Enhanced Chemical Vapor Deposition (PECVD), X-Section Sample Preparation, Doping, Diffusion, Atomic Layer Deposition (ALD), Sputter Deposition
- Clean-room/Operation:** Glove Box, Vacuum Systems Operation, Lab Inventory Management, Maintaining Safety Regulations, Clean-room Safety Documentation, Equipment troubleshooting, Tool Preparation
- Programming and Statistical Data Analysis:** Python, MATLAB, Excel, Design Of Experiments (DOE), JMP, Minitab
- Software:** COMSOL, Ansys HFSS, KLayout, Thermo-Calc, OriginPro, CrystalMaker, Figma, ImageJ

Leadership

STEP-UP Physics Mentor

Summer 2024

- Mentored 47 undergraduates, improved physics learning performance by 37% by incorporating flexible classroom teaching with curious examples, and by implementing the flipped classroom model.